

05/10/26

Signal Processing

- Recap: ROC properties, impact of stability, causality

- ### • Inverse z transform

- Inspection. Example: $X(z) = \left(\frac{1}{1 - \frac{1}{2}z^{-1}} \right)$, $|z| > \frac{1}{2}$. ✓

- Partial fraction method. Example: $X(z) = \frac{1}{(1 - \frac{1}{4}z^{-1}) \cdot (1 - \frac{1}{2}z^{-1})}$, $|z| > \frac{1}{2}$ ✓

Exampk: $X(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}$, $|z| > \frac{1}{2}$

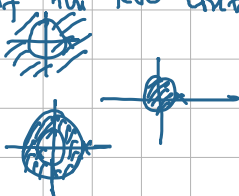
- Power Series expansion. Example: $X(z) = z^2(1 - \frac{1}{2}z^{-1})(1+z^{-1})(1-z^{-1})$. Pole at $z=0$.

Example: $X(z) = \frac{1}{1 - az^{-1}}$, $|z| > |a|$ Right-sided

Example: $X(z) = \frac{1}{1 - az^{-1}}$, $|z| < |a|$. Left-sided.

- Z-transform properties: Linearity, time-shift, multiplication by z^n , differentiation of $X(z)$, conjugation, time-reversal, convolution. ROC

- ROC properties:
 - Determined by circles in the z -plane
 - Poles are not contained in a ROC
 - DFT exists if the ROC contains the unit circle.
 - Right sided 
 - Left sided 
 - Two sided 



- Finite duration sequence: ROC is the entire z -plane, barring $0, \infty$ potentially.

- $X(z)$ has a pole at $z = -2$, $z = \frac{1}{2}$
 - three possible regions of convergence
 - $|z| > 2$; $|z| < \frac{1}{2}$; $\frac{1}{2} < |z| < 2$
 - causal
 - stable

- We cannot have this signal be causal and stable.

• Inverse z-transform: Inspection, partial fraction expansion, division/series expansion

• Inspection: $X(z) = \frac{1}{1-az^{-1}}$, $|z| > |a|$

$$\Rightarrow x[n] = a^n \cdot u[n] \quad (\text{by inspection of } X(z) \text{ + ROC})$$

• Partial fraction expansion: $X(z) = \frac{1}{(1-\frac{1}{4}z^{-1})(1-\frac{1}{2}z^{-1})}$, $|z| > \frac{1}{2}$

$$= \frac{2}{(1-\frac{1}{2}z^{-1})} - \frac{1}{(1-\frac{1}{4}z^{-1})}$$

$$\Rightarrow x[n] = 2 \cdot \left(\frac{1}{2}\right)^n u[n] - \left(\frac{1}{4}\right)^n u[n].$$

$$X(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}, \quad |z| > 1$$

$$= \frac{(1+z^{-1})^2}{(1-\frac{1}{2}z^{-1})(1-z^{-1})}$$

$$= \frac{A}{1-\frac{1}{2}z^{-1}} + \frac{B}{1-z^{-1}} + C$$

$$A = -9, \quad B = 8, \quad C = 2$$

$$\frac{-9}{1-\frac{1}{2}z^{-1}} + \frac{8}{1-z^{-1}} + 2 = \frac{2 \cdot (1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}) - 9 \cdot (1-z^{-1}) + 8(1-\frac{1}{2}z^{-1})}{(1-\frac{1}{2}z^{-1})(1-z^{-1})}$$

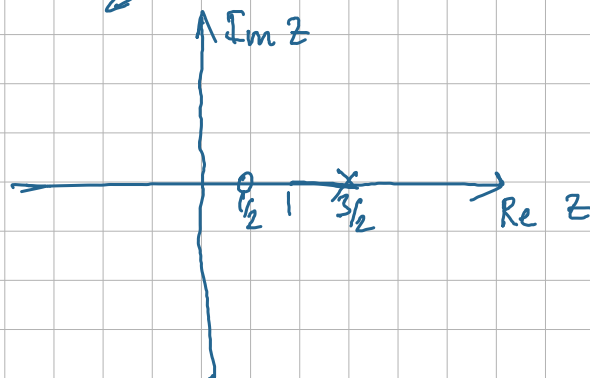
$$= \frac{z^{-2} - \cancel{3z^{-1}} + \cancel{2} - \cancel{9} + 9z^{-1} + \cancel{8} + 4z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - z^{-1})}$$

$$= \frac{z^{-2} + 2z^{-1} + 1}{(1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2})} \quad \checkmark$$

$$\Rightarrow x[n] = 2\delta[n] - 9\left(\frac{1}{2}\right)^n u[n] + 8u[n]$$

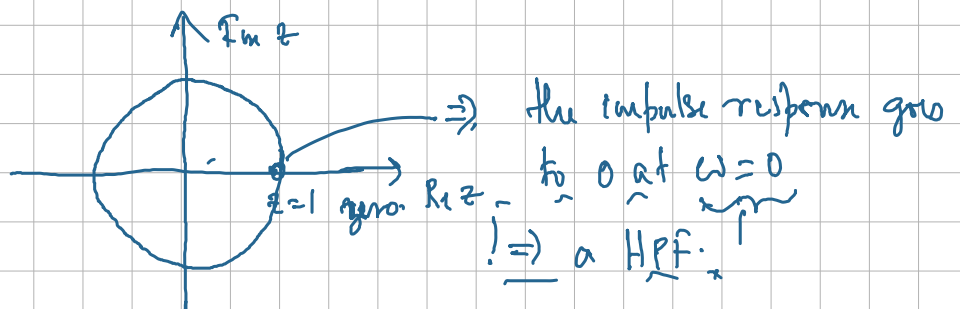
- Poles at $z = \frac{1}{2}$, $z = 1$
- Two zeros (repeated) at $z = -1$

Let's say $H(z) = \frac{(z - \frac{1}{2})}{(z - \frac{3}{2})}$, $|z| < \frac{3}{2}$ ↖ Check how this system behaves.



Simpler case: $H(z) = (z - 1)$ ROC is the entire z plane

$$\Rightarrow h[n] = \delta[n+1] - \delta[n] \quad (\text{a differencing non causal system})$$



- Series expansion for inverse z -transform:

$$X(z) = \frac{1}{1 - az^{-1}}, \quad |z| < |a|$$

Come up with a series expansion for this $X(z)$ in power of z .